INCH-POUND
MIL-PRF-49316A(CR)
27 AUGUST 1997
SUPERSEDING
MIL-T-49316(CR)
26 April 1991

PERFORMANCE SPECIFICATION
TEST SET, ELECTRONIC SYSTEMS, TS-3895A/UV

This performance specification is approved for use by USACECOM, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification establishes the performance, test, manufacture and acceptance requirements for the Test Set, Electronics System, TS-3895A/UV, herein referred to as the "Test Set".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.f). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-781	-Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of the above specification, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.f).

DRAWINGS

USA CECOM

5003414	- Case, Deep Drawn, Military
A3134090	- Control Panel Assembly
A3134092	- Identification Plate, Large
A3134097	- Identification Plate, Small
A3134099	- Case Assembly

(Copies of documents, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.f).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 122-89 - Sample Size to Estimate a Measure of Quality for a Lot or Process; choice of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC-S-815A - General Requirements for Soldering Electronic Interconnections

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2.d), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Test Set is a self-contained test set designed to check the Aviator's Night Vision Imaging System (ANVIS), AN/AVS-6(V)1 and AN/AVS-6(V)2 for proper operation. A darkened, sheltered area is required for proper use. A collimation attachment and a diopter scope are special tools which comprise an attachment kit for use with the Test Set. The Test Set also checks proper operation of the Night Vision Goggles (NVG) AN/PVS-5 (A, B, C) and AN/PVS-7 (A, B) as specified herein.

3.4 Design. The Test Set shall meet the form, fit, function and interface requirements as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.4.1 Dimensions. The test set shall meet the dimensions and dimensional tolerances of drawing 5003414 to ensure proper physical interface with existing hardware.

3.5 Performance characteristics.

3.5.1 ANVIS connections. The Test Set shall provide a mounting block and two electrical connections for the power pack. A simulated aircraft power connector shall mate to the power pack input connector and provide voltage to the power pack. This voltage shall simulate the power supplied by the power converter for ANVIS, which is mounted in the aircraft. The other electrical connections shall simulate the ANVIS load to monitor power pack current and voltage output. It also shall monitor the low battery indicator output.

3.5.2 Objective lenses interface. Each Test Set test port shall interface with the objective lenses listed in Table I. All objective lens interfaces shall be sufficiently light-tight to permit proper testing of the ANVIS or NVG in a darkened environment.

TABLE I. Objective lens maximum outside diameter.

System	Diameter
AN/PVS-5A	1.000 in.
AN/PVS-5B	1.355 in.
AN/PVS-6(V)1 and (V)2	1.355 in.
AN/PVS-7A	1.355 in.
AN/PVS-5C	1.455 in.
AN/PVS-7B	1.455 in.

3.5.3 Collimation attachment. The collimation attachment shall permit viewing through one monocular at a time or both simultaneously. When viewing both monoculars simultaneously, the two separate images shall be combined into one overlapped scene. To ensure proper image overlap, the collimation attachment shall have the two inputs collimated within 3 minutes or less in the vertical direction and 6 minutes or less in the horizontal direction. The horizontal direction is the axis connecting the two inputs to the collimation attachment (see 3.5.13).

3.5.4 Zero diopter scope. A low power telescope as described herein shall be easily attachable and detachable from the collimation attachment.

3.5.4.1 Diopter scope magnification. The diopter scope shall have a magnification of 5 X ± 1 X.

3.5.4.2 Diopter scope focus. The diopter scope focus shall be infinity, ± 0.4 diopter.

3.5.4.3 Diopter scope entrance aperture. The entrance aperture of the diopter scope shall be a minimum of 20mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.5 Collimator lenses. The collimator lenses used in the Test Set to project the test pattern shall be compatible with the ANVIS and NVG Objective lenses as specified herein.

3.5.5.1 Equivalent focal length (EFL). The EFL of the collimator lens shall be 66mm, ± 1mm.

3.5.5.2 Field of view (FOV). The target format covered by the Test Set collimator lens shall be sufficient to present a minimum 40 degree FOV to the ANVIS or NVG.

3.5.5.3 f/number. The f/number of the collimator lens shall be f/2.9 or less.

3.5.5.4 Relative illumination (RI). The RI shall be equal to or greater than 80 percent with a 23mm stop placed 11mm before the first collimator element as seen by the user over the calculated semifield angle subtended by the target format. The relative illumination shall beequal to or greater than 55 percent at 20 degrees semi-field angle for the same conditions as stated above.

3.5.5.5 Modulation transfer function (MTF). The collimator lens shall have on-axis MTF performance on-axis and at semi-field angles of 3.5 degrees and 7 degrees for both radial and tangentil orientations as listed in Table II over the spectral region defined in Figure 1 with a 23mm stop placed in front of the lens.

TABLE II. Test Set Collimator Lens MTF.

	3.5 degrees		7 degrees		
1p/mm	On-axis	Radial	Tangential	Radial	Tangential
0	100	100	100	100	100
5	75	75	75	71	63
10	68	65	63	46	35
15	63	58	50	14	27
20	57	48	37	8	20

3.5.5.6 Lens coatings. Lens coatings shall be optimized to match the spectral characteristics of the light source (see 3.5.6).

3.5.6 Light source. The light source shall be a Light Emitting Diode (LED) with a peak output at 820 nanometers ± 25 nanometers (see Figure 1).

3.5.7 Simulated aircraft power. The Test Set shall provide 3.8 Vdc at +0.2/-0.4 Vdc 150 mA for mating to the ANVIS power pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.8 ANVIS power pack output. The Test Set shall test the current and voltage output of an ANVIS power pack. Using GO/NO-GO visual indicators, the Test Set shall accept (GO)) a power pack that produces voltages between the lower and upper voltage linits listed in Table III and reject (NO-GO) a power pack that produces voltages outside these limits also listed in Table III.

TABLE III. Test Set power pack acceptance criteria.

Power pack voltage	Current draw
Lower Limit 2.4 Vdc, ± 0.10 VDC Upper Limit 3.1 Vdc, ± 0.10 VDC	73mA, ± 8mA 94mA, ± 10mA

3.5.9 System current. The Test Set shall test the current drawn by NVG systems listed in Table IV. A different binocular connector shall be used for testing each system. Other than connector selection, no other operator control is required to test the different system. The GO/NO-GO limit for system current is contained in Table IV. Operating voltage is 2.9Vdc, ± 0.2Vdc.

TABLE IV. GO limit for system current.

System	Upper current limit	Lower current limit
AN/AVS-6	67, ± 7mA	30, ± 6mA
AN/PVS-5	36, ±6mA	16, ±4mA
AN/PVS-7	20, ± 4mA	9, ± 3mA

NOTE: All values are measured at room temperature (see 3.12.2).

3.5.10 Low battery indicator. The Test Set shall test the voltage output of an ANVIS power pack when the power pack low battery indicator circuit is activated. Using GO and NO-GO visual indicators, the Test Set shall accept (GO) power packs that supply 1.3Vdc ± 0.05Vdc to 2.5 Vdc +/- 0.05 Vdc to the LED pin of the power pack connector and reject (NO-GO) those that do not. When the Test Set is in this test mode, the simulated aircraft power output to the power pack from the Test Set shall be reduced to 2.8, ± 0.3 VDC.

3.5.11 Resolution. The six patterns used for high light level and six patterns used for low light level testing shall have resolution as defined in Table V when projected by the Test Set collimator lens. The Test Set shall present a target (see Figure 2) to the ANVIS or NVG under test. The brightness of this target through the collimator lens shall be as specified in 3.5.12 Table VI. The target shall have a contrast of 98 percent or higher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

TABLE V. Pattern resolutions.

Pattern No.	High light	Low light
1	0.47 CY/MR, ± 7 percent.	0.19 CR/MR, ± 7 percent
2	0.53 CY/MR, ± 7 percent	0.21 CY/MR, ± 7 percent
3	0.59 CY/MR, ± 7 percent	0.23 CY/MR, ± 7 percent
4	0.67 CY/MR, ± 7 percent	0.26 CY/MR, ± 7 percent
5	0.75 CY/MR, ± 7 percent	0.30 CY/MR, ± 7 percent
6	0.84 CY/MR, ± 7 percent	0.33 CY/MR, ± 7 percent

NOTE: The two projected patterns shall be within 7 degrees of the center of the field of view. No light sources other than the test target shall be visible when the Test Set is in this mode.

3.5.12 Radiometric light output. With the Test Set passing the self test feature the average light level in the focal plane of a 27 ± 1mm EFL, T-1.58, maximum clear aperture of 23mm test objective lens mounted to the Test Set output port shall be as listed in Table VI. The reading shall apply over a 6mm diameter circle centered in the test objective focal plane. Table VI is based on 820 ± 25 nanometers radiation specified in 3.5.6. All values are watts/cm^2.

TABLE VI. Average light levels for test objective lens mounted to the Test Set output port.

AN/PVS-5/AN/PVS-7	AN/AVS-6(ANVIS)	Selector
7.89×10^{-9} ± 50 percent	2.14×10^{-9} ± 50 percent	High Light
3.09×10^{-11} ± 30 percent	8.37×10^{-12} ± 20 percent	Low Light

3.5.13 Collimation. The two patterns projected by the Test Set shall be collimated within 3 minutes or less as measured by the center circle of the test patterns. When combined with collimation attachment, the total collimation error shall be 9 minutes or less in the horizontal direction and 6 minutes or less in the vertical direction. The edges of the rectangles on the two patterns shall be aligned rotationally with each other within 2 degrees or less.

3.5.13.1 Collimation test pattern. The Test Set shall present a pattern to the binocular that consists of a high density black rectangle of 2 degrees ± 10 minutes width and 1 degree ± 10 minutes height (see Figure 2). Centered in the rectangle shall be a 6 minute ± 1 minute clear circle.

3.5.13.2 Focus. In both the high light level or low light level test mode, the test pattern shall be focused at infinity +0.0/-0.020 diopters for 820 nanometer illumination.

3.5.14 Vignetting. The Test Set shall present a full 40 degree minimum field of view test target to the ANVIS or NVG under test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.15 Weight. The Test Set shall weigh no more than 7.257 pounds (16 kilograms) including attachments and instrument case using a scale with an accuracy of 1% or better.

3.5.16 Size. The Test Set shall measure 50x40x35cm (l x w x h) to within ± 0.1 cm.

3.5.17 Controls. The Test Set shall include four controls: a Power Switch, a Function Selector Switch, GO/NO-GO indicators and a NVG/ANVIS selector knob.

3.5.17.1 Power switch. The Power Switch shall be a three position switch of ON-OFF-POWER CHECK. The power check position shall be a push-to-hold position that checks power to the system from any power source used and checks operation of the visual GO/NO-GO indicators.

3.5.17.2 Function Selector Switch. The Function Selector Switch shall set all the electronics and activate the GO/NO-GO indicators for the following tests:

Color	Function
Orange	Self Test (test light source brightness)
Black	Power pack
White	System Current
Blue	High Light Level, Resolution/collimation/Tube Stability
Green	Low Battery Indicator
Yellow	Low Light Level Resolution

NOTE: Positions of the function selector switch shall be marked with color coding as well as with words. Each position shall have a positive detent. The GO/NO-GO indicators shall be off in the High Light Level and low Light Level function positions.

3.5.17.3 GO/NO-GO indicators. The GO indicator shall be a green light and NO-GO indicator shall be a red light.

3.5.17.4 NVG/ANVIS selector knob. The function of the NVG/ANVIS selector knob shall be to change the radiometric output of the Test Set as specified in 3.5.12.

3.5.18 Electrical interface.

3.5.18.1 Power sources. The Test Set shall be capable of operating at 110 VAC, ± 10V (50-70 Hz), 220 VAC, ± 10V
(40-60 Hz), and 28 VDC, ± 6 VDC.

3.5.18.2 Power fuses. Each power source shall have its own line fuse. The AC line will use a 1/4 slo-blo fuse and the DC line will use a 1 AMP slo-blo fuse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.5.18.3 Polarity reversal. Components shall not be damaged when subjected to polarity reversal due to inadvertent reversal of the 28 VDC power. Component operation shall be normal when proper polarity is restored.

3.5.19 Environmental stress screening (ESS). Each Test Set shall be subjected to thermal cycling and random vibration as specified in Figure 3, followed by a 10 minutes vibration per Figure 4, to surface infant mortality (see 3.12.4).

3.6 Environmental conditions.

3.6.1 Temperature (operating). The Test Set with attachments shall be capable of continuous operation within the temperature range of -10°C to +52°C.

3.6.2 Temperature (non-operating). The Test Set with attachments shall be cable of withstanding storage and transit within the temperature range of -62°C to +85°C.

3.6.3 Altitude. The Test Set shall be cable of operation at altitudes up to 15,000 feet, and storage (non-operating) at altitudes up to 40,000 feet.

3.6.4 Humidity. The Test Set shall withstand the effects of humidity greater than 94 percent in both storage and operation over a temperature of +20°C to +30°C for operation and + 65°C for storage. Fogging on internal glass surfaces shall not occur and electronics shall not be damaged. Replacement of the desiccant during the test is allowed a maximum of four (4) times.

3.6.5 Vibration. The Test Set shall not be damaged after being subjected to vibration within the frequency range and amplitude shown in Table VII.

TABLE VII. Acceptance criteria.

Frequency increment	Vibration level (peak to peak)	Sweep time per increment	Survey time per axis	Dwell time per axis	G level at maximum frequency
5-15 Hz	1.5mm	7 min.			0.7
15-25 Hz	1.0mm	3 min.	15 min.	10 min.	1.3
25-55 Hz	0.5mm	5 min.			3.0

3.6.6 Mechanical shock pulse. The Test Set shall not be damaged (see 3.12.1) after being subjected to a 30 “g” half sine wave shock pulse on any face of the instrument case.

3.6.7 Bench handling. The Test Set shall not be damaged after being subjected to shocks normally associated with bench handling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.6.8 Bounce, loose cargo. The Test set with attachments installed shall be capable of withstanding the shocks and vibration associated with normal vehicular transportation that may be encountered during field employment, deployment, and logistic support.

3.6.9 Fungus. The Test Set shall not support the growth of or be damaged by exposure to fungus growth as encountered in tropical climates.

3.6.10 Electromagnetic interference. The Test Set shall comply with the electromagnetic interference requirements of Appendix A to this specification.

3.7 Interchangeability. Same type test sets, assemblies, subassemblies, special tools and replaceable parts shall be physically and functionally interchangeable without modification. Exceptions are as follows: focusing of the collimator lenses and adjustment of the LED monitor. Individual items shall not be hand picked for fit or performance.

3.8 Reliability. The Test Set shall have a minimum mean time between failure (MTBF) of 1000 hours when operated from 110 Vac, 220Vac or 28 Vdc power. Replacement of light bulbs, fuses, and desiccants shall not be considered reliability failures.

3.9 Treatment and painting.

3.9.1 Metals and metal treatments. All selected materials shall be of corrosion resistant types or shall be treated to resist corrosion. It is recommended that all metallic parts be treated and painted in accordance with drawing A3134100. Standard MS or AN parts do not require refinishing.

3.9.2 Finishes. U.S. Army olive drab is the preferred color unless precluded by operational requirements. All external surfaces, except light transmitting elements, that are part of the test light path shall be prepared in a non-reflective finish.

3.10 Nameplates and product marking.

3.10.1 Nameplates. Each Test Set shall be marked with a serialized nameplate on the control panel assembly. Nameplate information is identified on drawing A3134097. Recommended locations of data plates are shown on drawing A3134090.

3.10.2 Marking. Each Test Set shall be permanently and legibly marked for identification. Marking information is identified on drawing A3134092. Recommended locations of markings are shown on drawing A3134099.

3.11 Workmanship. All electronic parts, components, assemblies and sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

3.11.1 Soldering. Soldering shall be in accordance with ANSI/IPC-S-815A.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5 mandatory for this specification.

3.12.1 Damage. Damage is defined as any breakage, loosening, shifting, or failure of any finish, hardware, connection, component, or condensation on any internal surface, or failure to operate in accordance with the requirements of this specification.

3.12.2 Room temperature. Room temperature is defined as + 23°C, ± 10°C.

3.12.3 Infant mortality. Those failures which occur due to poor workmanship or weak components.

3.12.4 Environmental Stress Screening (ESS). An ESS failure is any shortcoming, imperfection or operational non-conformance, including a one-time non-repeatable anomaly, either sudden or gradual in nature, which causes the unit under test to deviate from specified limits. ESS failures, include poor workmanship (i.e. solder balls or cracks), defective manufacturing processes (i.e. wave soldering devices which cannot handle the heat) and defective components. Non-ESS failures are those failures directly attributed to improper installation in the test facility, failures of test monitoring equipment, failures due to operator errors, failures occurring during repair, and failures attributed to test equipment generating overstress conditions.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Visual inspection. Each first article shall be inspected in accordance with Table VIII, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that assembly and may be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

TABLE VIII. Inspection.

Inspection	Requirement paragraph	Test paragraph
Construction	3.4	Visual/gauges
Treatment and painting not as specified	3.9	4.9
Marking not as specified	3.10.2	4.9
Workmanship not as specified	3.11	4.9
ESS	3.5.19	4.5.19

4.2.2 Tests. Following successful completion of the inspections specified in 4.2.1, each of the first article Test Sets shall be subjected to the tests of subgroup 1 of Table IX. Upon completion of subgroup 1 tests, a specified sample of Test Sets shall be subjected to the tests of Subgroup 2 and Subgroup 3 of Table IX. At least one Test Set will be subjected to tests for Fungus, EMI and the balance of the tests. A specified number of Test Sets shall be subjected to the subgroup 4 requirements of Table IX. Failure of any Test Set shall constitute rejection of that Test Set and may be cause for rejection of the first article. Refer to the contract or purchase order for sampling requirements (see 6.2.d).

4.2.2.1 Test for attachments. As separate items, the attachments shall be tested in accordance with Table IX, unless specified as part of the Test Set test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

TABLE IX. First article tests.

Test	Requirement paragraph	Test paragraph
Collimation attachment	3.5.3	4.5.3
Zero diopter scope magnification	3.5.4.1	4.5.4.1
Focus	3.5.4.2	4.5.4.2
Entrance Aperture	3.5.4.3	4.5.4.3
Operational and environmental Subgroup 1		
Equivalent Focal Length	3.5.5.1	4.5.5.1
Field of view	3.5.5.2	4.5.5.2
f-number	3.5.5.3	4.5.5.3
Relative Illumination	3.5.5.4	4.5.5.4
MTF	3.5.5.5	4.5.5.5
Light source	3.5.6	4.5.6
Simulated Aircraft Power	3.5.7	4.5.7
Power pack output	3.5.8	4.5.8
System current	3.5.9	4.5.9
Low battery indicator	3.5.10	4.5.10
Resolution	3.5.11	4.5.11
Radiometric light output	3.5.12	4.5.12
Collimation	3.5.13	4.5.13/4.5.13.1
Focus	3.5.13.2	4.5.13.2
Vignetting	3.5.14	4.5.14
Weight	3.5.15	4.5.15
Size	3.5.16	4.5.16
Controls	3.5.17	4.5.17
Electrical interface	3.5.18	4.5.18
Subgroup 2		
Temperature (Operating)	3.6.1	4.6.1
Temperature (Non-operating)	3.6.2	4.6.
Transmission	3.5.5.7	4.5.5.7
Altitude	3.6.3	4.6.3
Humidity	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Mechanical shock	3.6.6	4.6.6
Interchangeability	3.7	Visual
Subgroup 3		
Bench handling	3.6.7	4.6.7
Bounce, loose cargo	3.6.8	4.6.8
EMI	3.6.10	4.6.10
Fungus	3.6.9	4.6.9
Subgroup 4		
Reliability	3.8	4.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.3 Conformance inspection.

4.3.1 Inspection. Each Test Set and/or attachments shall be inspected in accordance with Table VIII, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that assembly or attachment.

4.3.2 Group A tests. Each Test Set, and/or attachments shall be tested in accordance with Table X. Each Test Set shall be subjected to and pass ESS specified in 3.5.19. Tests listed in Table X may be performed in any order. Failure of a Test Set or attachment to pass these tests shall be cause for rejection of that Test Set and/or applicable attachment.

TABLE X. Group A tests.

Test	Requirement paragraph	Test paragraph
Radiometric Light Output	3.5.12	4.5.12
Collimation	3.5.13	4.5.13
Focus	3.5.13.2	4.5.13.2
Controls	3.5.17	4.5.17
Collimation Attachment	3.5.3	4.5.3
Zero Diopter Scope	3.5.4	4.5.4
Magnification	3.5.4.1	4.5.4.1
Focus	3.5.4.2	4.5.4.2
Test Set Collimator Lens	3.5.5	4.5.5
EFL	3.5.5.1	4.5.5.1
Field of View	3.5.5.2	4.5.5.2
f/number	3.5.5.3	4.5.5.3
Relative Illumination	3.5.5.4	4.5.5.4
Transmission	3.5.5.7	4.5.5.7
MTF	3.5.5.5	4.5.5.5

4.3.2.1 Lot formation. Upon successful completion of Group A testing, Test Sets shall be formed into lots. The lot size shall be specified in the contract or purchase order (see 6.2).

4.3.3 Group B test. Test Sets which have successfully passed Group A tests shall be sampled and subjected to the Tests prescribed in Table XI. Sampling shall be in accordance with the contract or purchase order (see 6.2.g).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

TABLE XI. Group B tests.

Test	Requirement paragraph	Test paragraph
Light source	3.5.6	4.5.6
Simulated aircraft power	3.5.7	4.5.7
Power pack output	3.5.8	4.5.8
System current	3.5.9	4.5.9
Low battery indicator	3.5.10	4.5.10
Vignetting	3.5.14	4.5.14

4.3.4 Group C test. Group C testing shall be conducted on the Test Sets, with attachments where specified (if attachments are required with that lot), selected from units which have passed Group A tests. The Test Sets shall be tested in accordance with the tests listed in Table XII. Samples shall be selected in accordance with the contract or purchase order (see 6.2.g). Tests may be performed in any order.

TABLE XII. Group C tests.

Test	Requirement paragraph	Test paragraph
Subgroup 1		
Temperature (operating)	3.6.1	4.6.1
Temperature (Non-operating)	3.6.2	4.6.2
Vibration	3.6.5	4.6.5
Subgroup 2		
Humidity	3.6.4	4.6.4

4.3.4.1 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2.b).

4.3.4.2 Reinspection of conforming Group C sample unit. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection and test.

4.3.5 Group D test. At least four Test Sets shall be selected at random for tests in Table IX, Subgroup 4, every 180 days of production. The Test Set attachments shall not be subjected to this reliability test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.4 Examination. Each Test Set shall be examined for compliance with the requirements specified in 3.3 through 3.11. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection. The contractor/supplier may use his own test methods, measuring devices, tools, fixtures, and other equipment during the conduct of verification testing. This specification cites no test process or procedure as long as all the performances requirements per Section 3 are fully verified as being met.

4.5.1 Test conditions. Unless otherwise specified herein, the assembly shall meet the specified performance requirements at an ambient temperature of +23°C, ± 5°C. All specified tolerances are absolute values and errors due to test equipment shall be subtracted from those tolerances.

4.5.2 Performance tests.

4.5.2.1 Optical components. All optical components and lenses shall be examined for compliance with the requirements contained in Section 3. Use of applicable portions of 6.3 or equivalent methods are recommended.

4.5.2.2 Mechanical/Electrical components. Each Test Set shall be tested to verify compliance with the performance characteristics specified using the testing procedures contained in Section 4, herein. Inability for the Test Set to operate properly during such testing or failure of the Test Set to perform each performance characteristic shall constitute a failure as defined herein.

4.5.3 Collimation attachment. Place a flat mirror at least 4 inches in diameter perpendicular to the autocollimator which has an angular graduation at it's focal plane. Adjust the autocollimator perpendicular to the mirror. Place the collimation attachment between the autocollimator and the flat mirror so that the ray from the autocollimator goes through the outer port of the collimator port and splits into two paths to make the parallel beam. Those parallel beams will be reflected at the mirror back to the collimation attachment, then to the autocollimator. Separation of the double images represents the collimator error. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.3.1 Collimation attachment (alternate). Place a pinhole in the focal plane of a four inch or larger diameter collimator. The pinhole shall subtend three minutes or less of arc. Place the collimation attachment looking into the collimator with the shutter set in the "both" position. Look through the collimator attachment port with a low power telescope. The two pinholes must overlap. Failure of the pinholes to overlap shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.4 Zero diopter scope.

4.5.4.1 Diopter scope magnification. The magnification of the diopter scope shall be measured. Use of the procedure in 6.3 is recommended. Failure to meet the requirements of 3.5.4.1 shall constitute failure of this test.

4.5.4.2 Diopter scope focus. The diopter scope focus shall be measured with a diopter scope of at least power 3. Failure to meet the requirements of 3.5.4.2 shall constitute failure of this test.

4.5.4.3 Diopter scope entrance aperture. Measure the entrance pupil of the diopter scope. Use of the procedure in 6.3 is recommended. Failure to meet the requirements of 3.5.4.3 shall constitute failure of this test.

4.5.5 Collimator lenses.

4.5.5.1 Equivalent focal length (EFL). The equivalent focal length of the collimator lens shall be measured. Use of the procedure in 6.3 is recommended. A measuring microscope shall be used in place of a photographic plate. Failure to meet the requirements of 3.5.5.1 shall constitute failure of this test.

4.5.5.2 Field of view. The field of view of the collimator lens shall be measured. Use of the procedure in 6.3 is recommended. The format size shall be set 49.2mm. Failure to meet the requirements of 3.5.5.2 shall constitute failure of this test.

4.5.5.3 f/number. The f/number of the collimator lens shall be calculated by dividing the equivalent focal length measured in 4.5.5.1 by the effective aperture. The effective aperture of the collimator lens shall be measured. A recommended procedure for measuring the effective aperture is contained in 6.3. Failure to meet the requirements of 3.5.5.3 shall constitute failure of this test.

4.5.5.4 Relative illumination. The relative illumination shall be measured with a set-up that consists of a collimated light source, goniometer, and an integrating sphere incorporating a detector. The light source shall have the same spectral distribution as the test set (TS-3895A/UV) source. The relative signal ratio as a function of angle to the axial signal is the relative illumination. Failure to meet the requirements of 3.5.5.4 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.5.5 Modulation transfer function (MTF). The collimator lens with a 23mm stop shall be tested on an MTF apparatus that yields sine wave MTF response. The MTF apparatus shall have a light source and detector whose combination shall yield a spectral response curve complying with Figure 1. The MTF apparatus shall allow the collimator to be tested either continuously or at a minimum of 4 distinct, evenly distributed points from 0 to 20 1p/mm. The area analyzed shall be at least 0.3mm in diameter. Measure the MTF of the collimator lens on-axis and at semi-field angles of 3.5 degrees and 7 degrees in he same focal plane as the axial focus for both radial and tangential orientations. Failure to meet the requirements of 3.5.5.5 shall constitute failure of this test.The calibration procedure to be used is to mesure the MTF of a known size target (e.g. slit or pinhole). All references to size ar in the imge plane. The MTF of a pinhole is known mathematiclly as a first order Bessel Function. MTF=(2J1 (x))/x where x=3.14156 *(1p/mm) * (pinhole diameter in mm). The MTF of a pinhole goes to zero at a spatial frequency 1.22 times its reciprocical diameter. The MTF of a slit is known mathematically as (sine (x))/x where x=3.14156 *(1p/mm) *(slit width). the units for x are in radians. the MTF of a slit goes to zero at a spatial frequency at its reciprocical width. The procedure shall be to measure the MTF of the lens and then measure the MTF of tyhe lens with a very large pinhole/slit. In the case of microprocessor based MTF apparatus wich corrects for the pinhole/slit size, the size of the pinhole/slit entered into the computer software, when measuring the lens with large pinhole/ slit should not be greater than 1/40 of the large pinhole/slit. The geometrical size of the aerial pinhole image shall be 0.030mm +/- 0.001. the geometrical size of the slit shall be 0.025 +/- 0.001. The MTF of the large pinhole/slit can be determined by taking the ratio of the MTF curves of the lens with large pinhole/slit, divided by the MTF of the lens under test. Differences in Modulation of greater than three percent will invalidate the MTF results. See Figure 7 for MTF as a function of spatial frequency and pinhole diameter/slit width.

4.5.5.6 Lens coatings. Measure compliance with the requirement of 3.5.5.6 using 3.5.6 light source characteristics (Figure 1).

4.5.5.7 Transmission. Place the unit under test in a collimated ray bundle that spectrally is in complince with Figure 1. Place a stop of 23mm diameter 11mm from the first optical surface. Using an integrating sphere with detector of bandwith in accordance with Figure 8, measure the signl S1. Remove the lens and measure energy coming through the aperture nd note as S2. The transmission is S1/Ss2. Failure to meet the requirements of 3.5.3.7 shall constitute failure of this test.

4.5.6 Light source. Measure the irradiance from the LED or at the output of the Test Set with a silicon detector or equivalent (R1). Repeat the measurement with the same detector having a bandpass filter placed between the LED and the detector (R2). The bandpass filter shall be calibrated for transmission and shall have a bandpass region of greater than 790 nm and less than 850 nm measured at its half maximum point. If the filter's bandpass transmission is T, as measured at the peak LED wavelength, then the requirement of 3.5.6 is met if R2 = T x R1 ±10 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.7 Simulated aircraft power. Measure the voltage output from the simulated aircraft power connector when the selector switch is in the power pack position and a variable resistive load has been adjusted to draw current specified by 3.5.7. Failure to meet the requirements of 3.5.7 shall constitute failure of this test.

4.5.8 ANVIS Power pack test. Apply a variable voltage to the power pack connector. Vary the voltage over a range that exceeds those values listed in 3.5.8. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.5.9 System current test. Apply variable simulated resistive load to the various goggle connectors in the system "current mode" position. Vary the load over a range that exceeds those values listed in 3.5.9. Failure to meet the requirements of 3.5.9 shall constitute failure of this test.

4.5.10 Low battery indicator test. Place the selector switch in low battery indicator test position and apply a variable simulated voltage to the power pack connector. Vary the simulated voltage over a range that exceeds the value (1.3 Vdc, ± 0.3 Vdc) listed in 3.5.10. Also, with the function selector switch in this test mode, measure the voltage output from the Simulated Aircraft Power connector. Failure to meet the requirements of 3.5.10 shall constitute failure of this test.

4.5.11 Resolution test. From piece part inspection data obtain the resolution values (LP/mm) for all twelve resolution targets shown on the drawing. Multiply them by EFL measured per and divide by 1000 to obtain the required resolution. Compare with the requirements of 3.5.11. Failure to meet these requirements shall constitute failure of this test.

4.5.12 Radiometric light output. The light level exiting the test ports of the Test Set shall be measured as stated in this paragraph and shall be within the limits listed in Table VI. The light output shall be measured in the focal plane of a 27mm, ±1mm EFL, T-1.58, clear aperture less than 23mm. If the T-No is different, results shall be scaled to a t-1.58 value for comparison to Table VI. Instrumentation used to measure the focal plans irradiance shall be sensitive between 760-880 nanometers. Active area of the detector shall be limited to a 6mm diameter circle.

4.5.13 Collimation. With the reticle lens subassembly removed from the Test Set main plate, use supplemental illumination to light both test patterns. Place a supplemental long focal length lens of at least 100mm aperture so it collects light from both test ports. Place a screen in the infinity focal plane of this supplemental lens so that the image of the high light level resolution pattern from each port is seen on the screen. The center white dots from the two patterns shall overlap. Failure of the two white to overlap shall constitute failure of this test.

4.5.13.1 Collimation (alternate). Attach the collimation attachment set and diopter scope to the Test Set. Set the selector switch on HIGH LIGHT Resolution. Pull the NVG/ANVIS selector knob out. Illuminate the reticle pattern by shining a white light source through the hole on the left of the collimation attachment. Two white dots shall be overlapped or separated by less than one half of a dot size. Failure of the two white dots to overlap shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.13.2 Focus. The set shall be positioned in a holding fixture. A calibrated auto collimator with a light source emitting at the wavelength of the test set light source shall be directed toward the aperture of the test set collimator. The collimator on the test set shall be moved away from the test pattern and adjusted back toward the pattern for maximum collimator signal. Prior to using the autocollimator the focus of the electronic autocollimator is adjusted against a 1/20 wave optical flat. Failure to meet the requirements of 3.5.13.2 shall constitute failure of this test.

4.5.14 Vignetting.

4.5.14.1 Vignetting test (preferred method). Construct a special viewing device that consists of an image intensifier tube with a fiber optic faceplate and an objective lens of short enough focal length that more than a 40 degree FOV is imaged on the tube input. Affix an opaque annular ring reticle pattern to the fiber optic faceplate such that the inner radius represents 35 ± 3 degrees FOV. An eyepiece of any power may be used to view the image tube screen, provided the entire screen is visible. Look into either port of the Test Set to view the high light level resolution pattern. Position of the special viewer so one can see the edge of the high light level pattern simultaneously around the outer radius of the annular reticle pattern. Repeat for the other port. Failure to see the reticle pattern in either port outside the 40 degree annular ring constitutes failure of this test.

4.5.14.2 Vignetting Test (alternative method). Use one monocular of either a AN/AVS-6 or AN/PVS-5 that has been measured to have a minimum 40 degree field of view. Look into either port of the Test Set to view high light level pattern. Position the binocular so one van see the entire high light level pattern. Check to make sure none of the pattern's dark edges fall withoin the system's field of view. Repeat for the other port. If, for either port, a position cannot be found such that no dark edges appear, the Test set fails this requirement.

4.5.15 Weight. The Test Set and attachments shall be weighed on a scale accurate to within 10 grams. Failure to meet the requirements of 3.5.15 constitutes failure of this test.

4.5.16 Size. The Test Set with cover shall not exceed the dimensions as specified in 3.5.16. Failure to meet the requirements of 3.5.16 shall constitute failure of this test.

4.5.17 Controls.

4.5.17.1 Power switch. The power switch shall be examined for compliance with the requirements of 3.5.17.1 for both AC and DC operation. Failure to meet the requirements of 3.5.17.1 shall constitute failure of this test.

4.5.17.2 Function selector switch. The function selector switch shall be examined for compliance with the requirements of 3.5.17.2. Failure to meet the requirements of 3.5.17.2 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.5.17.3 GO/NO-GO indicators. When the power switch is held in the "power check" position, failure of either the GO (green light) or NO-GO (red light) indicator to illuminate constitutes failure of this test.

4.5.17.4 NVG/ANVIS Selector knob. The NVG/ANVIS selector knob shall be operated during testing for compliance with 3.5.12 and shall be examined for compliance with the requirements of 3.5.17.4. Failure to comply with 3.5.17.4 or properly support compliance with 3.5.12 shall constitute failure of this test.

4.5.18 Electrical interface.

4.5.18.1 Power sources. Test procedures 4.5.7, 4.5.8, 4.5.9, 4.5.10, 4.5.17.1 and 4.5.17.3 shall be repeated using 110 VAC ± 10 VAC at 60 Hz ± 10Hz, 220 VAC ± 10 VAC at 50 Hz ± 10 Hz and 28 Vdc, ± 6 Vdc. In addition, both AC and DC power shall be applied for 30 seconds simultaneously. The Test Set shall then be inspected for damage. Failure to meet the requirements of 3.5.7, 3.5.8, 3.5.9, 3.5.10, 3.5.17.3, or 3.5.18.1 for all AC and DC modes of operation shall constitute failure of this test.

4.5.18.2 Power fuses. The Test Set shall be inspected to insure fuses as specified in 3.5.18.2 are installed. Failure to meet this requirement shall constitute failure of this test.

4.5.18.3 Polarity reversal. Apply 28 Vdc power to the Test Set and turn the power switch on. Reverse the polarity of the 28 Vdc power source and apply to the Test Set for 30 seconds. Check the Test Set for damage. Failure to meet the requirements of 3.5.18.3 shall constitute test failure.

4.5.19 Environmental stress screening (ESS). Each Test Set shall be subjected to the thermal/vibration profile as shown in Figure 3. The profile shall consist of pre-system performance test followed by a ten minute vibration in accordance with Figure 4, followed by ten temperature cycles in accordance with Figure 3 and post-system performance tests. The following performance tests shall be conducted prior to and upon completion of ESS: Simulated aircraft power, power pack test, system current test, low battery indicator test, radiometric light output, collimation, focus, vignetting and controls. The unit will also be checked for workmanship and damage. During temperature cycling, equipment ON is defined as "Green Light" ON in SELF TEST mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.6 Environmental tests. Environmental testing may be performed in accordance with the methods and procedures identified in 6.4 The contractor may develop and use an alternative, (equivalent) test plan following Government review and approval. When the contractor's test plan deviates from the recommended test procedures, the contractor shall be required to show where his own tests and methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing of all environmental requirements shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar equipment for which the technical capabilities to properly simulate extreme climatic or environmental conditions are scientifically documented.

4.6.1 Temperature (operational). Verify compliance with the requirement of 3.6.1. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.2 Temperature (non-operational). Verify compliance with the requirement of 3.6.2. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.3 Altitude. Verify compliance with the requirement of 3.6.3. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.4 Humidity. Verify compliance with the requirement of 3.6.4. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.5 Vibration. Verify compliance with the requirement of 3.6.5. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.6 Mechanical shock pulse. Verify compliance with the requirement of 3.6.6. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.7 Bench handling. Verify compliance with the requirement of 3.6.7. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.8 Bounce loose cargo. Verify compliance with the requirement of 3.6.8. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

4.6.9 Fungus. Verify compliance with the requirement of 3.6.9. The tests recommended in 6.4, or an equivalent method may be used. Failure to meet the requirement shall constitute failure of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

4.6.10 Environmental (Non-operating) tests. Operational test and visual examination shall be conducted prior to and after each group conformance test. The collimation of the Test Set shall measured only at room temperature for environmental tests.

4.6.11 Operational environmental tests. When operation of the Test Set is required during the conduct of testing, the operation and performance of the testing shall be as defined in Table XIII. The tests shall be performed at the environmental extremes.

TABLE XIII. Operational tests during environmental.

Test	Requirement paragraph	Test paragraph
Power pack output	3.5.8	4.5.8
System current	3.5.9	4.5.9
Low battery indicator test	3.5.10	4.5.10
Power switch	3.5.17.1	4.5.17.1
G/NO-GO Indicators	3.5.17.3	4.5.17.3

4.7 Accuracy of testing apparatus. Except for environmental test apparatus, all other test instruments and test equipment used by the contractor shall have an allowable error consistent with the variable to be measured.

4.8 Reliability. The Test Set shall be tested in accordance with the contractor's own prepared Government approved test plan. Recommend use of MIL-HDBK-781, Test Plan XVC under "Combined Environments for Fixed Ground Equipment" of paragraph 50.1, Appendix B except as modified herein.

a. Temperature cycling between operating and storage temperatures shall be as shown in figure 5.

b. Each unit shall be subjected to ESS of prior to start of the reliability test.

c. All voltage stress shall be based on 110 VAC nominal.

d. Performance checks (Group A Tests) will be per Table XI and Table X with the exception of the attachment tests and the collimator lens tests.

e. Perform random vibration as specified in Figure 6 on 20 percent of Test Sets per test cycle, in rotating lots, insuring that all Test Sets are vibrated at least six times.

4.9 Workmanship and marking. Prior to assembly into the unit (s); assemblies modules and printed circuit boards shall be examined for compliance with this specification. All external movable controls shall be operated mechanically. Assembled units shall be examined for compliance with equipment specifications, workmanship and marking requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.h). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Test Set is intended for use in military applications for the maintenance and support of the AN/AVS-6, Aviator's Night Vision Imaging System (ANVIS) as well as to support the AN/PVS-5 and AN/PVS-7, Night Vision Goggles by reading identifying whether or not the image intensifier tubes are flight worthy.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this performance specification.

b. Necessary actions by contractor in the event of a first article failure or Group C failure.

c. MIL-STD-462 may be used for measuring electromagnetic interference.

d. Quantity and schedule for first article testing for all line items.

f. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

g. Sampling for Group B and Group C testing. As guidance, unless otherwise specified sampling should be conducted per the requirements of ASTM E 122-89.

h. Packaging requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

6.3 Design and test requirements for optical lens. The design and test requirements for lenses contained in MIL-STD-150, and listed below, are recommended, but are not mandatory for compliance with the performance requirements of this specification. The contractor may use any commercial design and test method that will assure compliance with the stated performance requirement in Section 3.

Inspection	Specification paragraph	MIL-STD-150 paragraph	MIL-STD-150 page
Equivalent Focal Length	3.5.5.1	3.1.4	3
(test for)		5.1.2.2	21
Field of View	3.5.5.2	3.5	7
f-number	3.5.5.3	4.1.3.3	18
Relative illumination	3.5.5.4	3.6.7	12
Transmittance		3.6.8	14
(test for)		5.1.2.10	24
Veiling glare		3.6.10	15
Focal length	3.5.5.1	4.1.5	19
Resolution	3.5.11	3.6.2	8
(test for)		5.1.1.7	20
Collimation	3.5.13	5.1.1.1	19

6.4 Environmental inspection procedures. The test methods and procedures of MIL-STD-810, listed below, are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

Inspection	Spec. paragraph	MIL-STD-810 method	MIL-STD-810 procedure	Remarks
Temperature (operating)	3.6.1	501.1/502	I	Operating temp is -10°C for 6 hours and +52°C for 6 hours.
Temperature (non-operating)	3.6.2	501.1/502	I	Non-operating temp. is -62°C for 6 hours and +85°C for 6 hours.
Altitude	3.6.3	500	I	Operate altitude chamber from 15 k to 40 k for 3 hours each.
Humidity	3.6.4	507.1	II	Follow 3.6.4 for a 48 hour cycle.
Vibration	3.6.5	516.2	II	Use Table VII for test.
Mechanical shock pulse	3.6.6	516.2	II	Use 30 G @ 4 milliseconds duration.
Bench handling	3.6.7	516	V	
Bounce, loose cargo	3.6.8	514.2.	XI	
Fungus	3.6.9	508.1	I	

6.5 Definitions. (See 3.12)

6.6 Subject term (key word) listing.

ANVIS
Night Vision
GO/NO GO Testing

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR
(Project 5805-0104)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

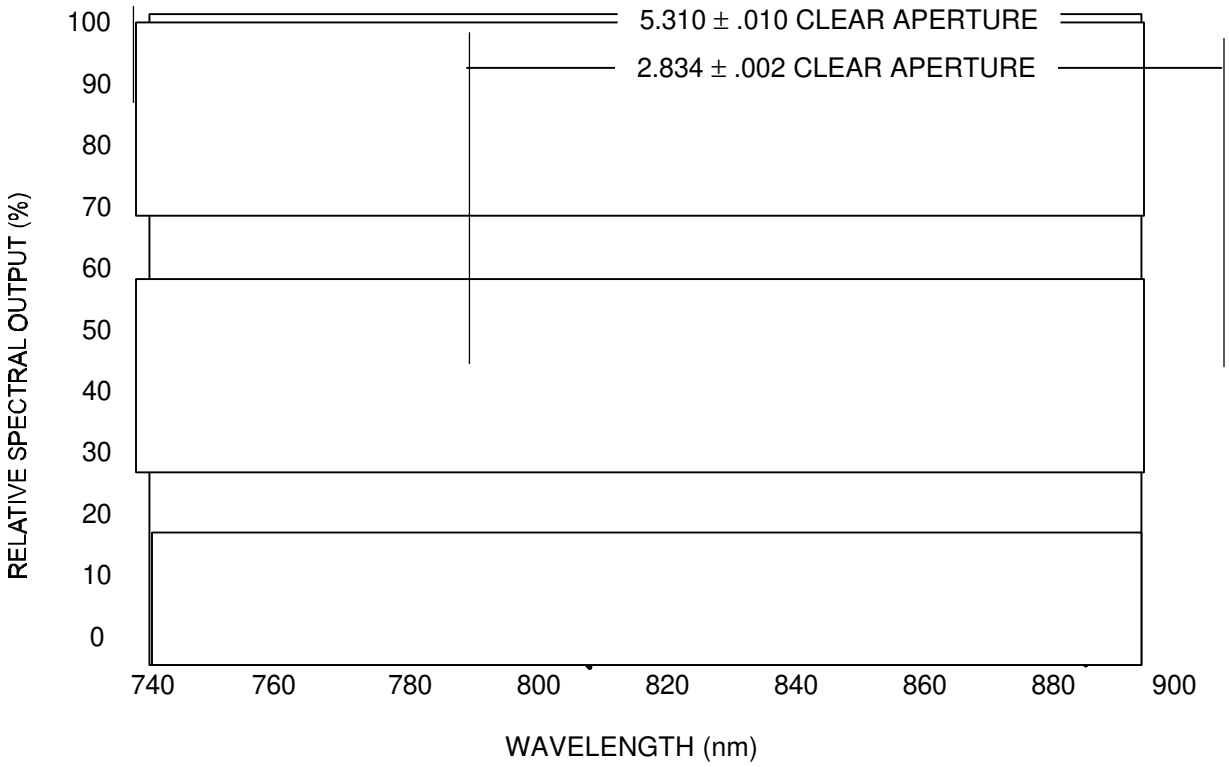


FIGURE 1. Typical spectral output versus wavelength.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

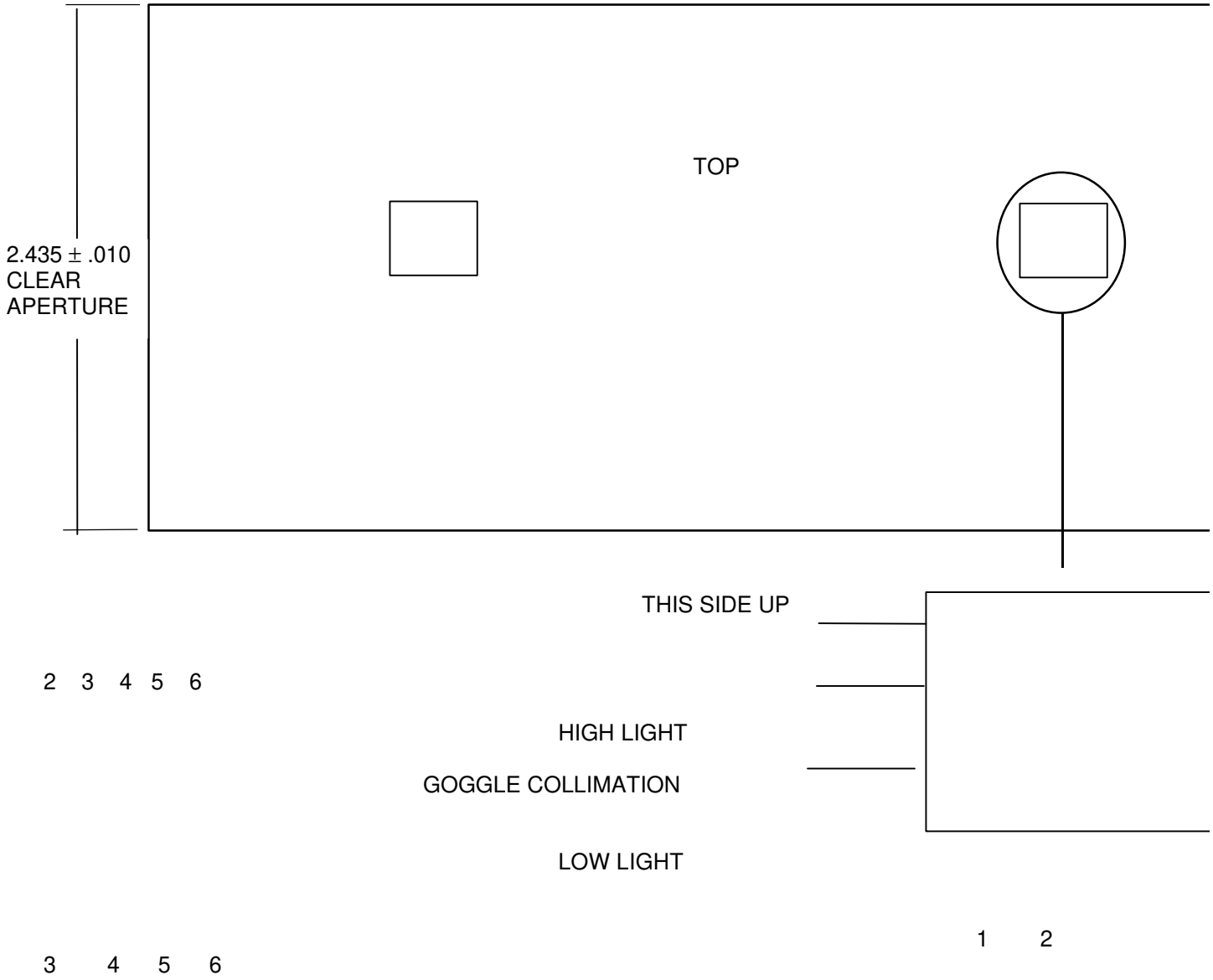


FIGURE 2. Single reticle pattern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

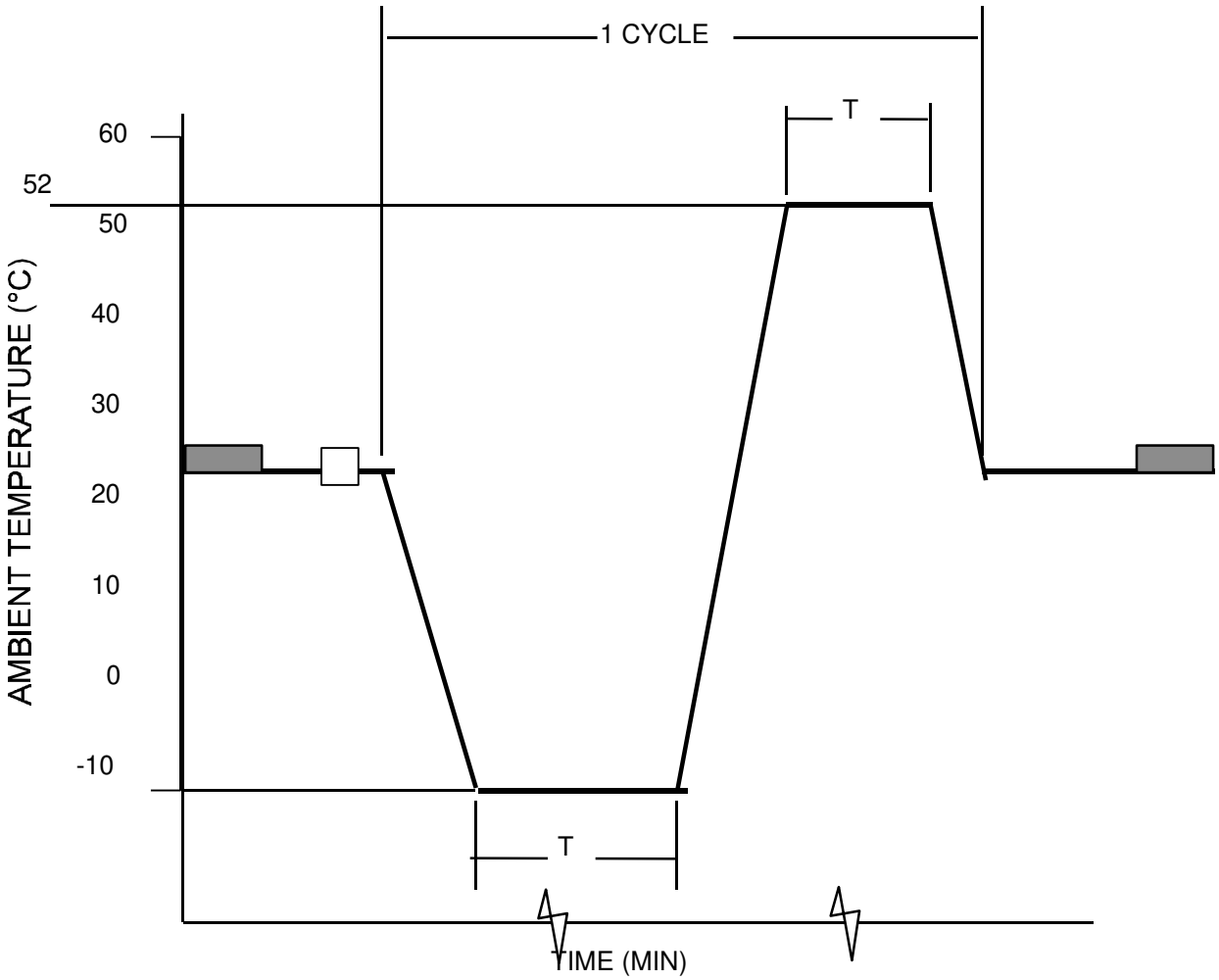


FIGURE 3. Thermal /vibration cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

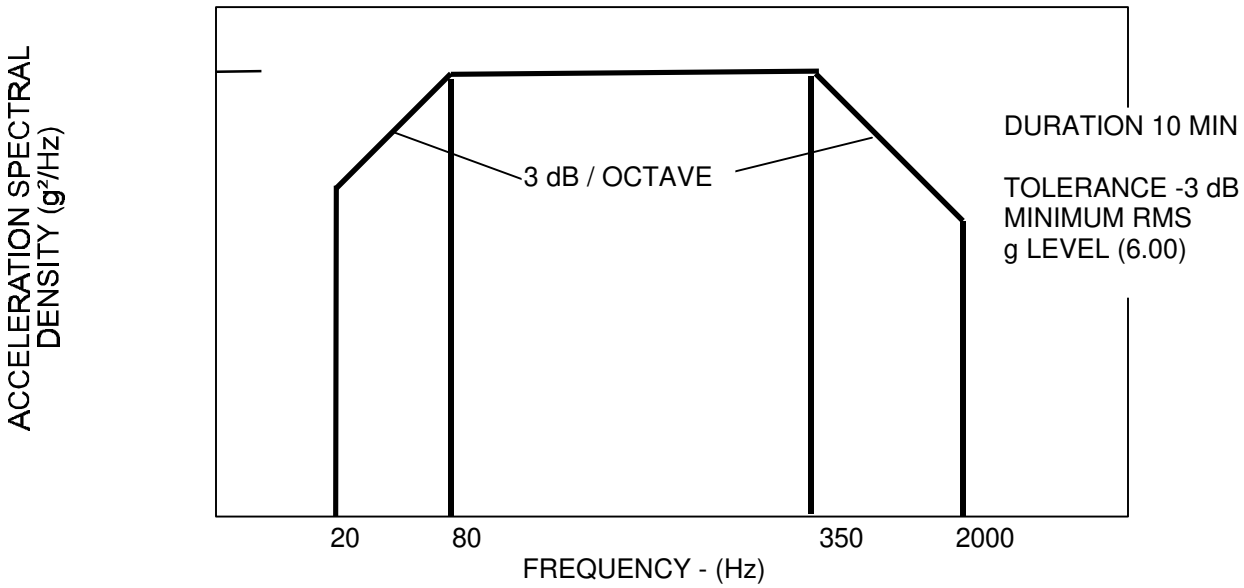


FIGURE 4. Random vibration acceptance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

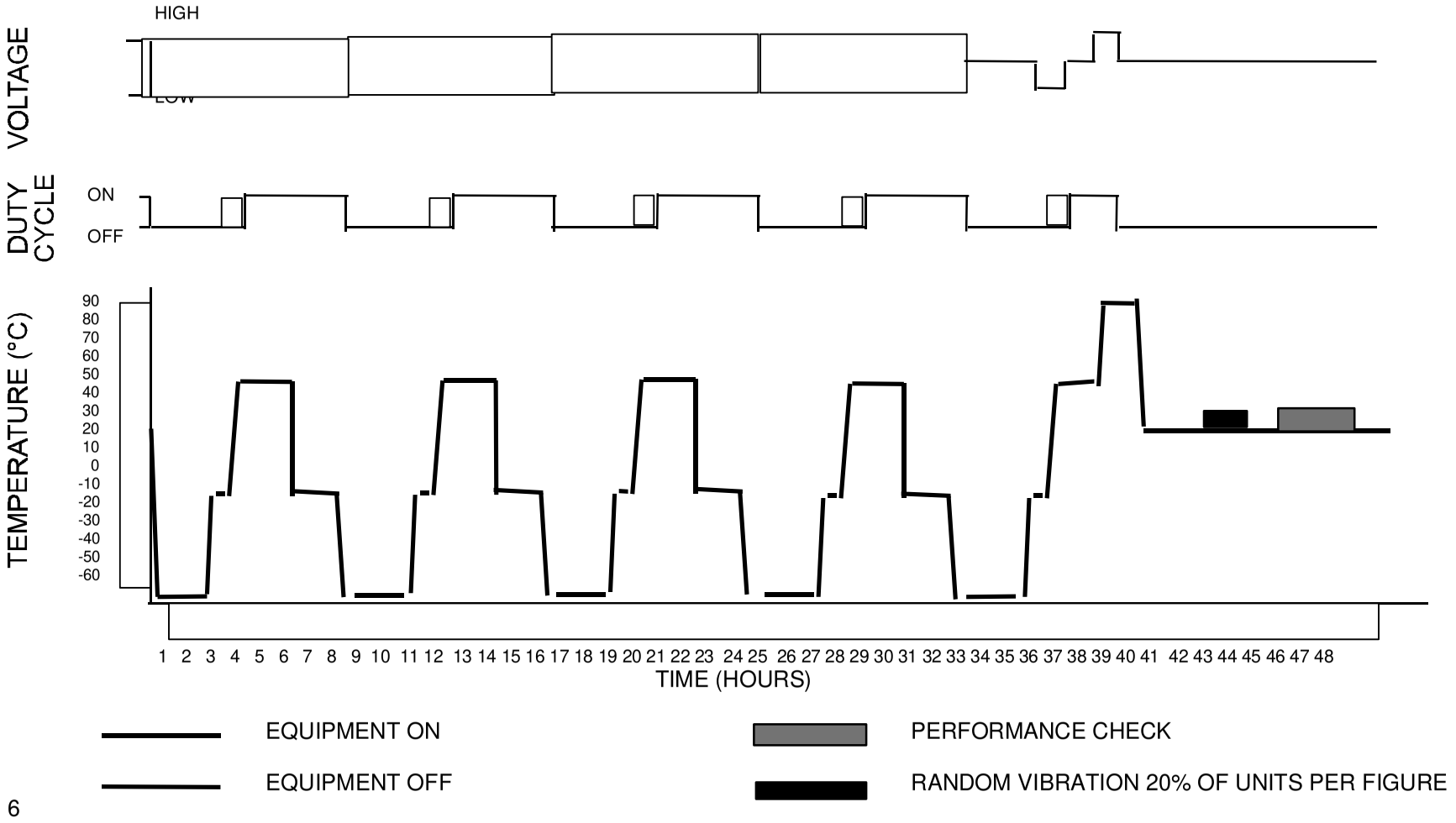


FIGURE 5. Temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

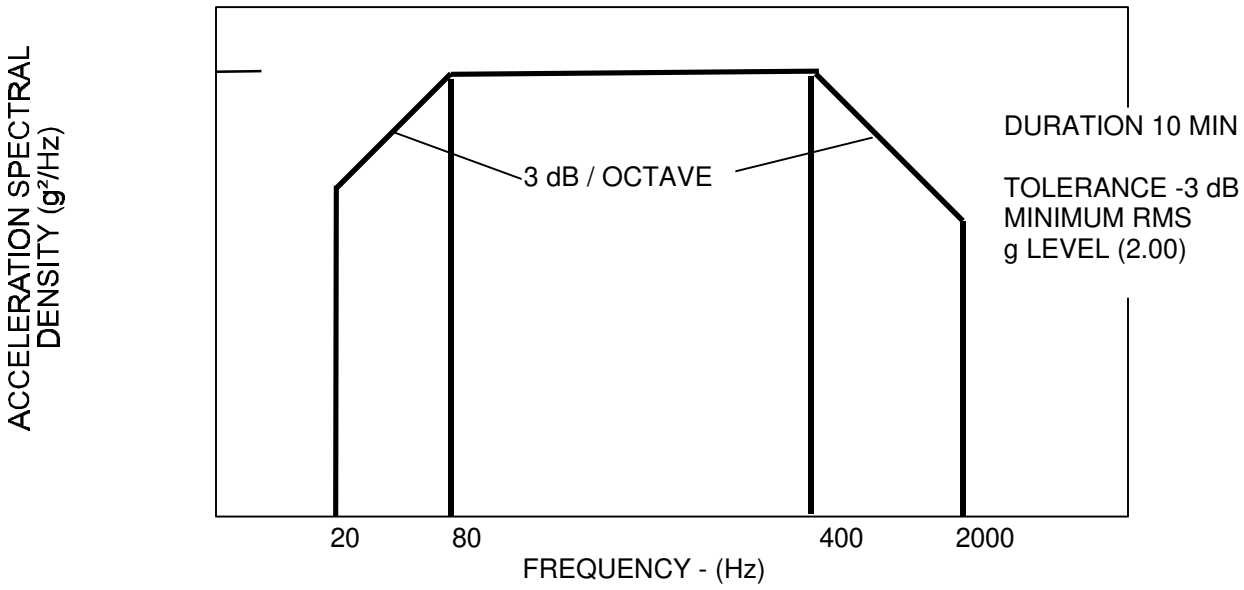


FIGURE 6. Random vibration spectrum (reliability).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A

ELECTROMAGNETIC INTERFERENCE (EMI) VERIFICATION PROGRAM FOR TEST SET, ELECTRONIC SYSTEMS, TS-3895A/UV

A.1 SCOPE

A.1.1 Scope. This appendix outlines and specifies the requirements for the electromagnetic interference (EMI) verification program applicable to the Test Set, Electronics System, TS-3895A/UV, herein referred to as the "Test Set". It is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 The EMI verification program for the Test Set shall be completed as per the following test plans:

a. EMI Qualification Test Plan, to be performed on first article equipment as specified in the contract or order.

b. EMI Production Verification Test Plan, to be performed on samples of production equipment as specified in the contract or order.

A.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

A.3 TEST CONDITIONS

A.3.1 The equipment shall be set up in a shielded enclosure with the test equipment and its related antennas and sensing equipment.

A.3.2 The test shall be conducted at normal room ambient temperature of 20° C to 30°C (68°F to 86°F).

A.3.3 The equipment under test shall be accessible to the operator for switching and retuning as required.

A.4 VERIFICATION PROCEDURE

A.4.1 Conducted emissions, power leads, 10 KHz to 10 MHz. The equipment is to be tested for conducted emissions on the power leads over the frequency range of 10 KHz to 10 MHz using the measurement setup of Figure A-1. Tests shall be performed at three frequencies each in the AM mode, 5.7 MHz, 39 MHz and 72.3 MHz. Equipment is to be tested in the transmit mode only. The emissions shall not exceed the levels shown in Figure A-2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

A.4.2 Conducted emissions, antenna terminal. The equipment is to be tested for conducted emissions on the antenna terminal over the frequency range hereby defined using the measurement setup of Figure A-3. The start frequency shall be as follows:

Operating frequency range	Start frequency of test
10 KHz to 3 MHz	10 KHz
3 MHz to 300 MHz	100 KHz

The conducted emissions at the antenna terminal of the equipment under test shall not exceed the values below.

a. Receivers: 34 dBµV

b. Transmitters (standby mode): 34 dBµV

c. Transmitters (transmit mode): Harmonics, except the second and third, and all other spurious emissions shall be at least 80 dB down from the level at the fundamental. The second and third harmonics shall be suppressed 50 + 10 log p (where p = peak power input in watts at the fundamental) or 80 dB, whichever requires less suppression.

A.4.3 Conducted susceptibility, antenna port, intermodulation, 15 KHz to 10 GHz. The equipment is to be tested for conducted susceptibility to intermodulation on the antenna port over the frequency range of 15 KHz to 10 GHz using the measurement setup of Figure A-4. Tests shall be performed at the same three frequencies in both the FM and SSB mode. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in paragraph 3.5.10 of this specification, when subjected to the test signals specified in the equipment specification.

A.4.4 Conducted susceptibility, antenna port, rejection of undesired signals, 30 Hz to 20 GHz. The equipment is to be tested for conducted susceptibility to undesired signals on the antenna port over the frequency range of 30 Hz to 20 GHz using the measurement setup of Figure A-4. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in paragraph 3.5.9 and 3.5.11 of this specification when subjected to the test signals specified in the equipment specification.

A.4.5 Radiated emissions, electric field, 2 MHz to 18 GHz. The equipment is to be tested for radiated emissions of the electric field over the frequency range of 2 MHz to 18 GHz using the measurement setup of Figure A-5. Narrowband emission limits shall be as indicated on Figure A-6. However, the transmit fundamental frequency is exempt from this requirement. Additionally, a maximum of ten emissions, harmonically related to the transmit fundamental, and the 11.0 MHz oscillator may exceed the Figure A-6 limit by 20 dB per test frequency and one other emission may exceed the prescribed limit by no more than 33 dB. The Broadband limit below 150 Khz is

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

relaxed 10 dB due to the 50 Khz oscillator. The emissions shall not exceed the levels shown in Figure A-6. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields.

A.4.6 Radiated susceptibility, electric field, 10 KHz to 40 GHz. The equipment is to be tested for radiated susceptibility to the electric field over the frequency range of 10 KHz to 40 GHz using the measurement setup of Figure A-7. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the electric fields shown below in Table A-I. Up to 30 MHz, the limits shall be met for vertically polarized fields. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields. Circular polarized fields are not acceptable.

Table A-I. Radiated susceptibility limits - electric field.

Frequency range	Limit level (volts/meter)
10 KHz-2 MHz	20
2 MHz - 30 MHz	50
30 MHz - 1 GHz	50
1 GHz - 18 GHz	50
18 GHz - 40 GHz	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A

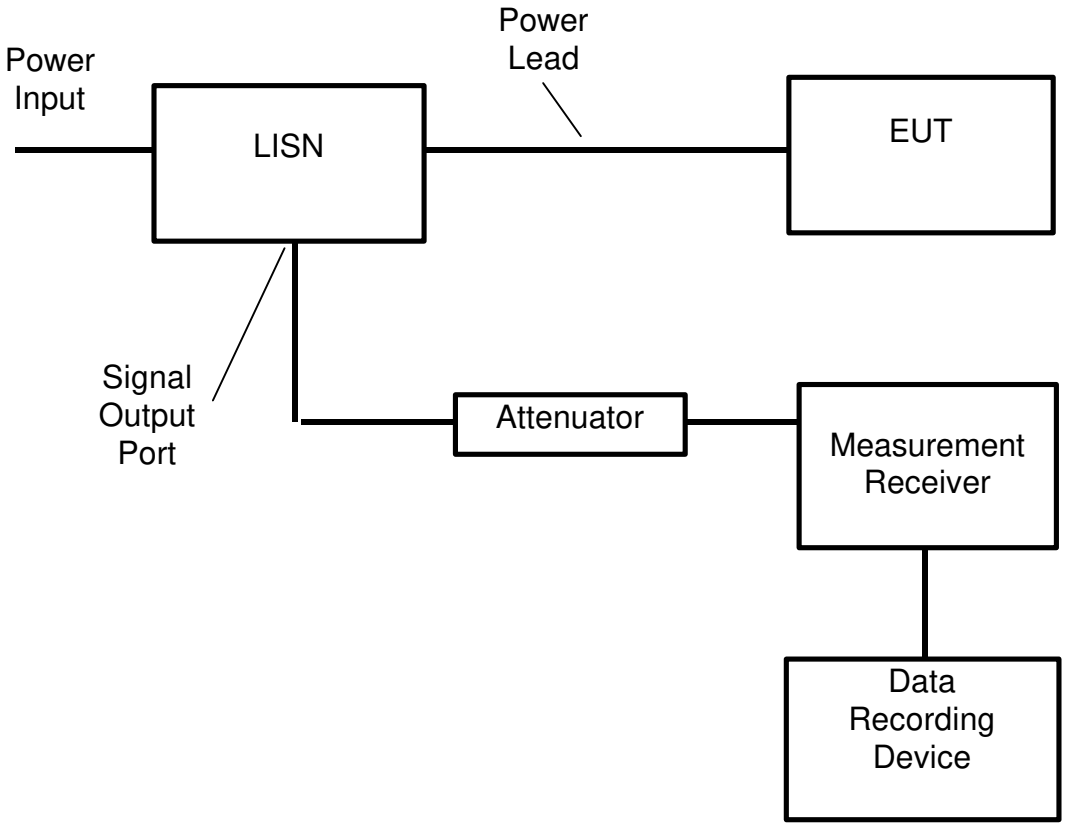
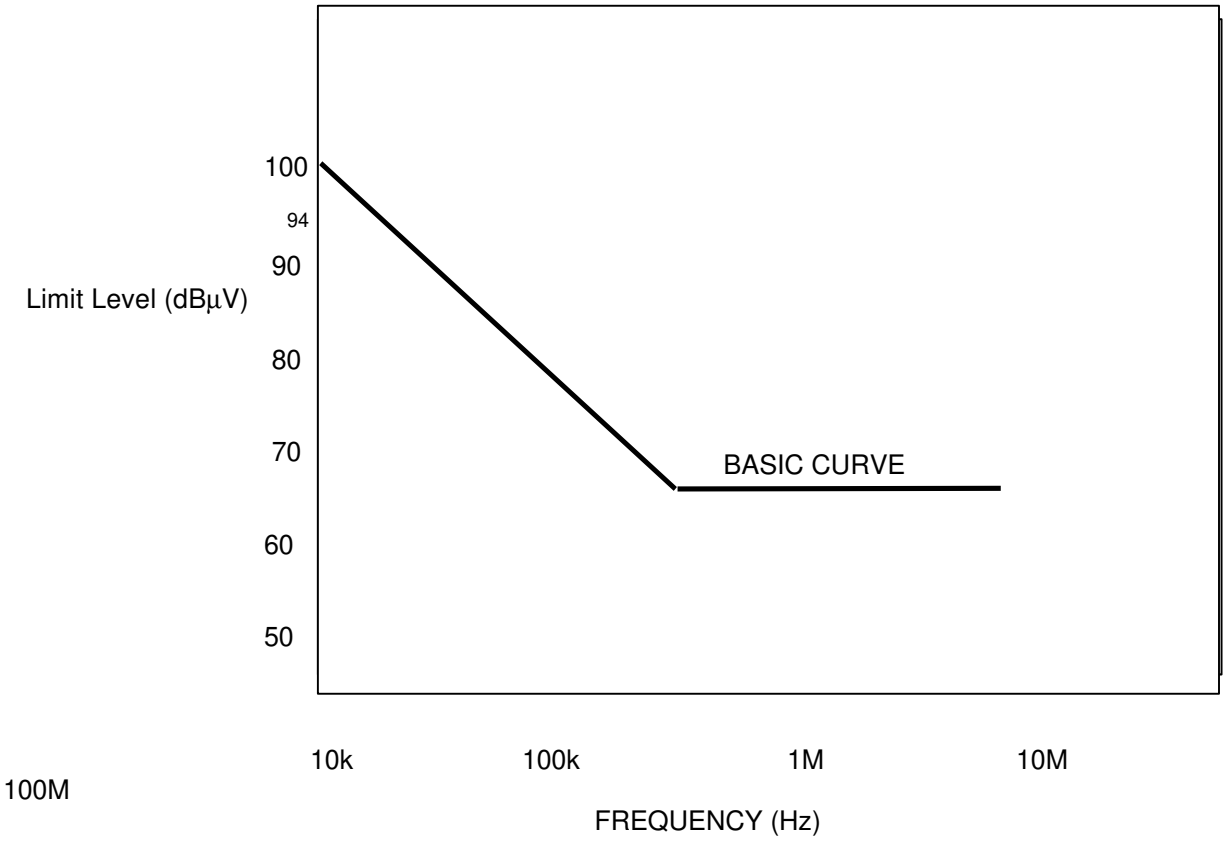


FIGURE A-1. Measurement setup, conducted emissions, power leads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.



NOMINAL EUT SOURCE VOLTAGE (AC&DC)	LIMIT RELAXATION BASIC CURVE
28 V	6dB
115V	9dB
220V	10dB
270V	12dB
440V	12dB

Figure A-2. Conducted Emissions limit, power leads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

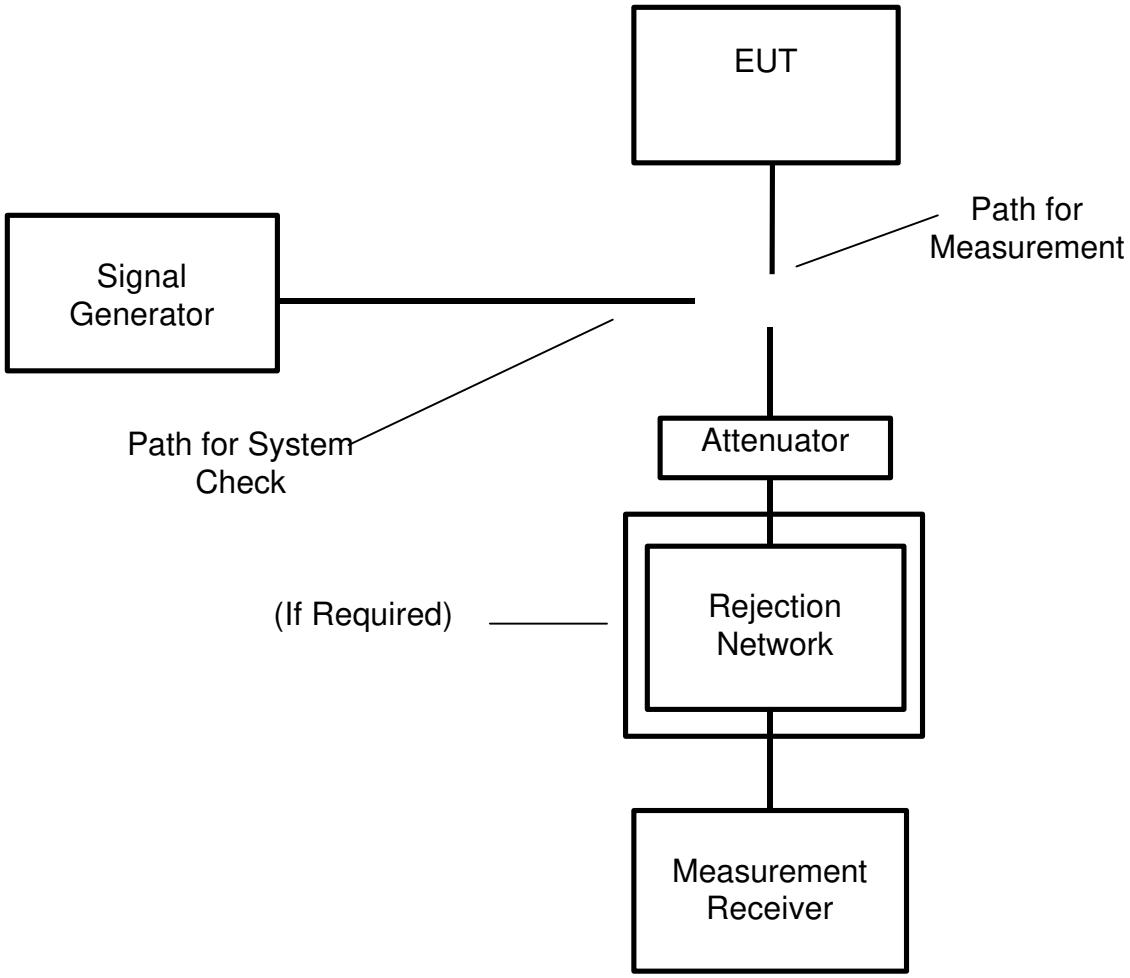
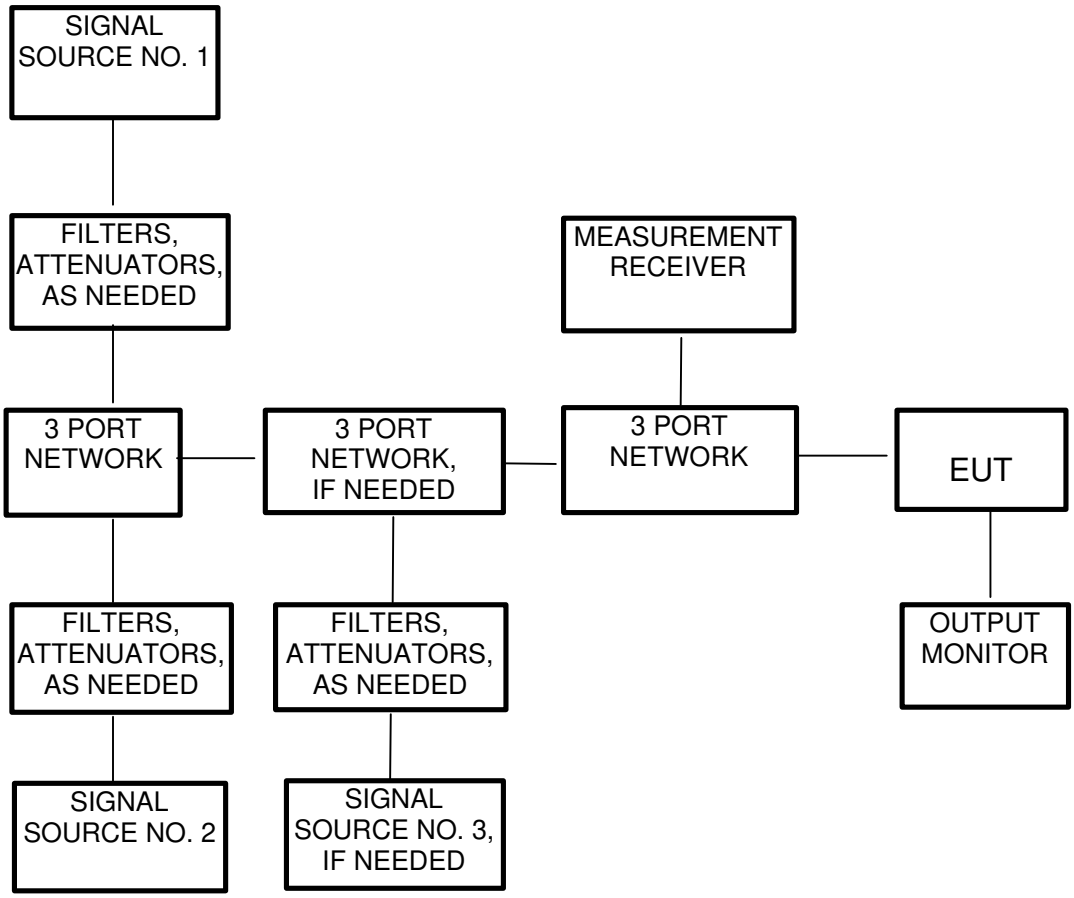


FIGURE A-3. Measurement setup, conducted emissions, antenna terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A



IGURE A-4. Test setup, conducted susceptibility, antenna port.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

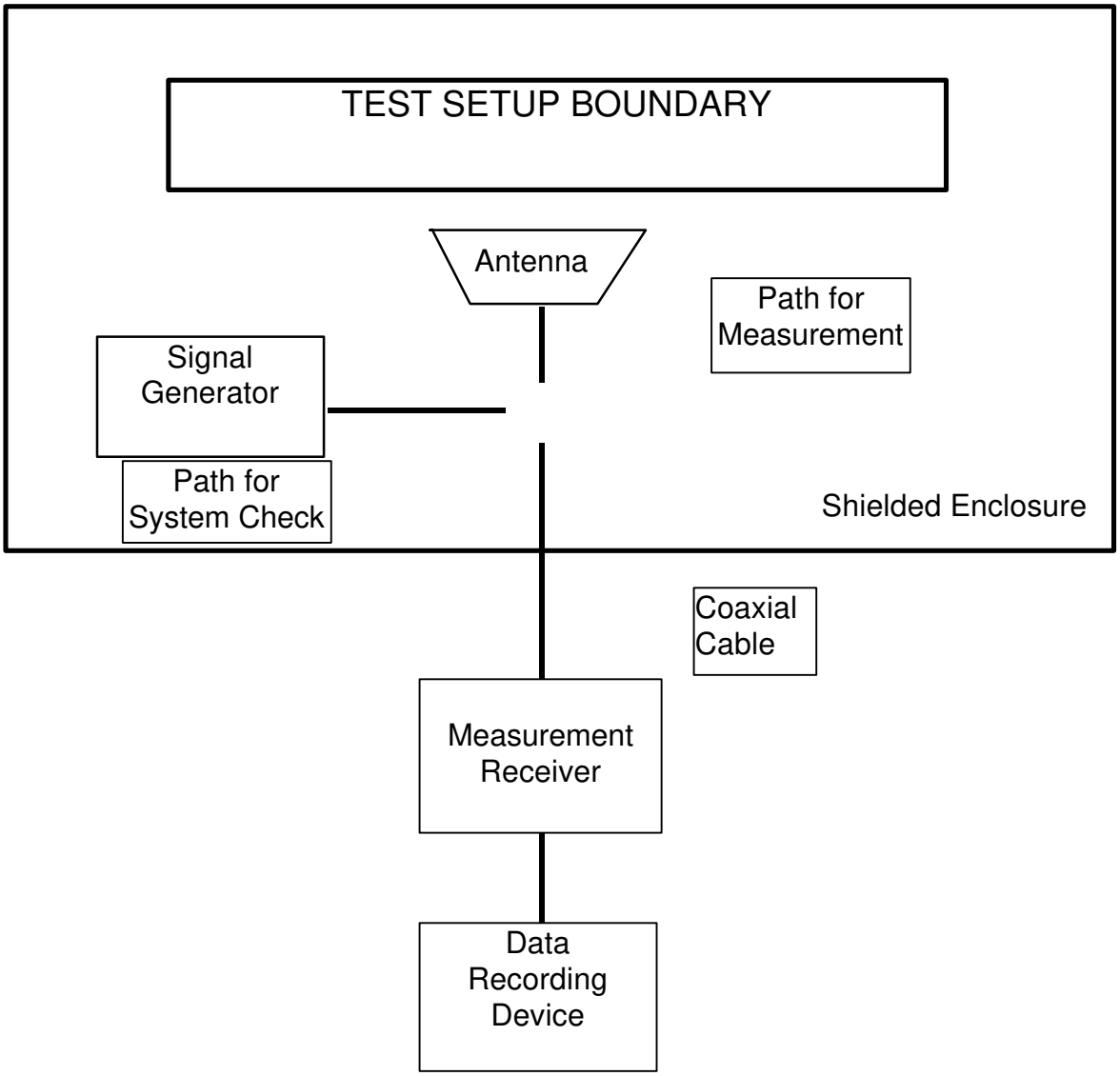


FIGURE A-5. Test setup, radiated emissions, electric field, 2 MHz to 18 GHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

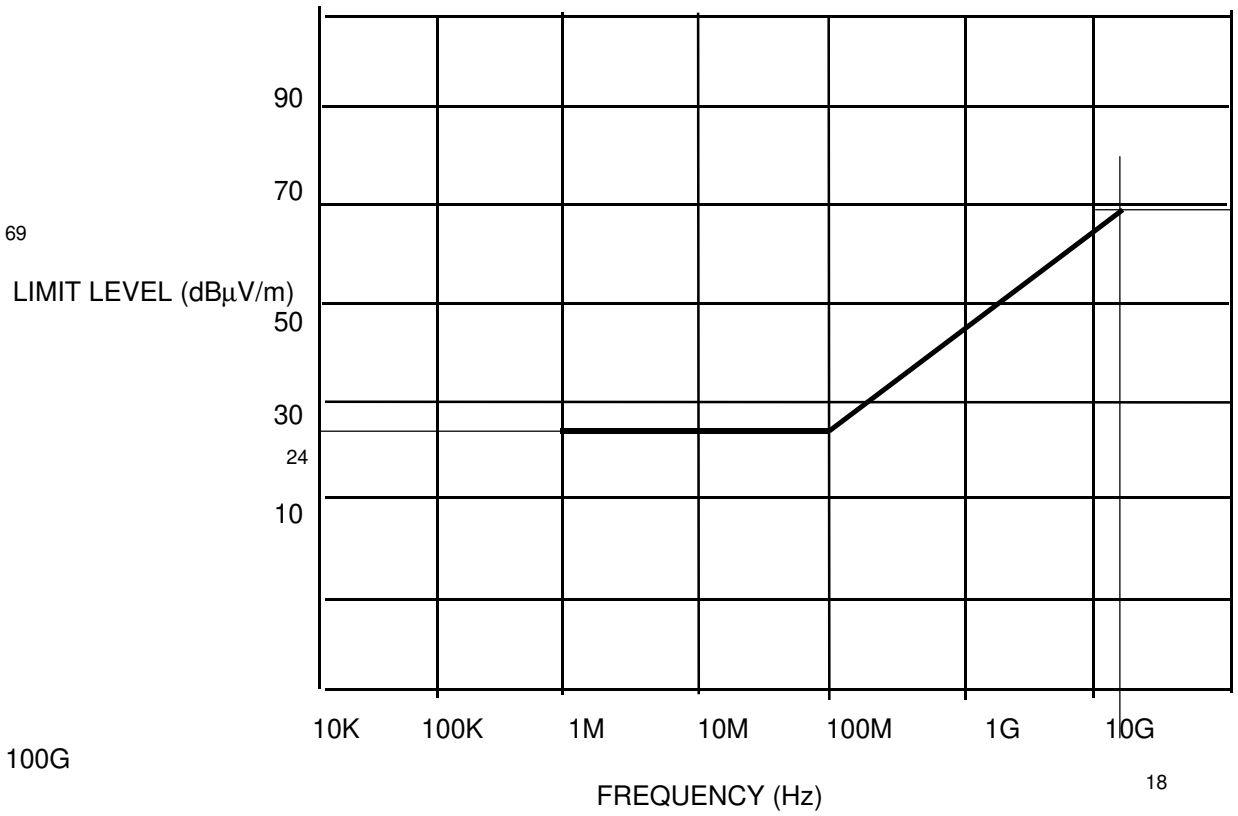


Figure A-6. Radiated Emission limit, electric field.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

APPENDIX A

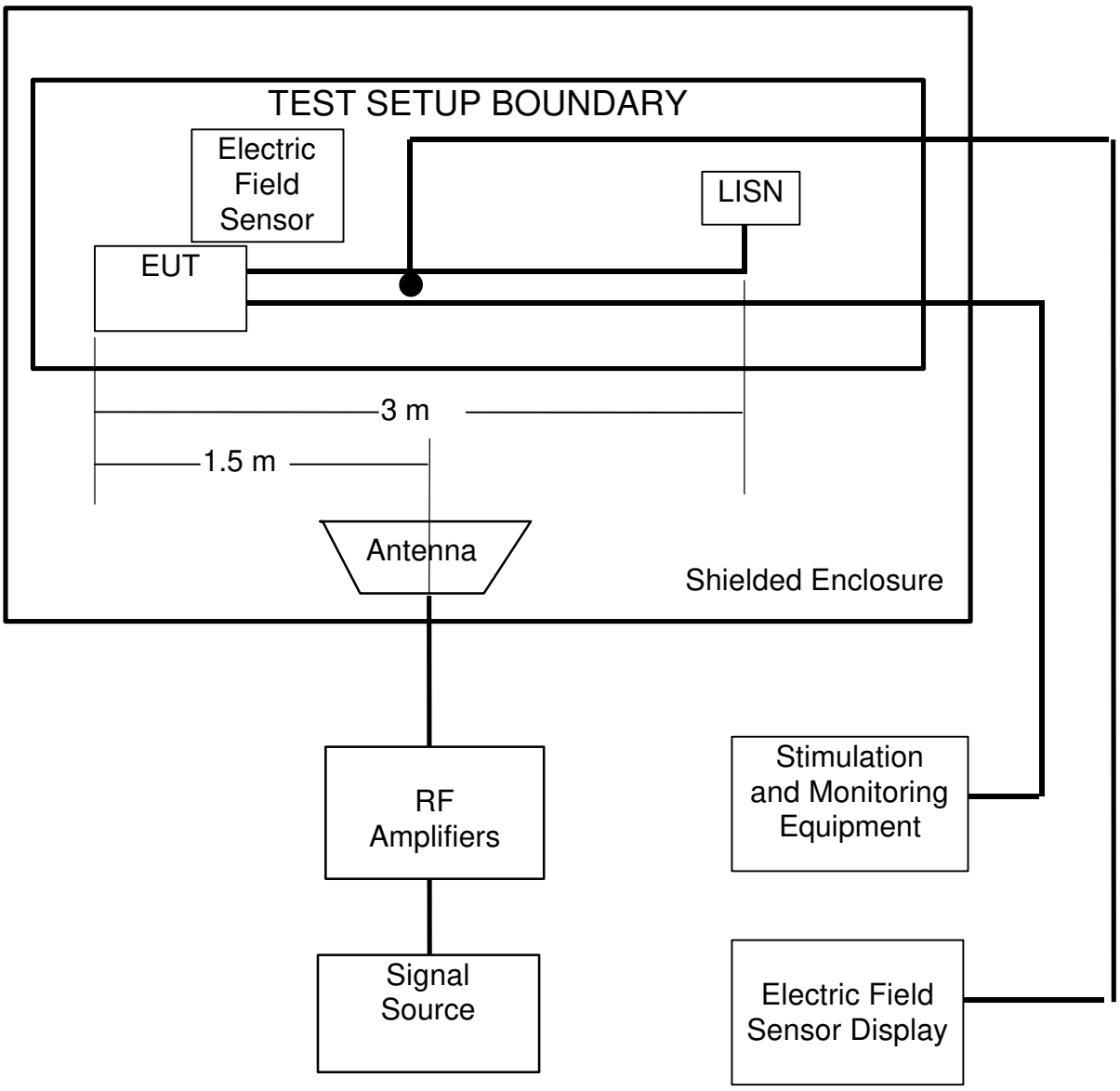


FIGURE A-7. Test setup, radiated susceptibility, electric field, 2 MHz to 40 Ghz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49316	**2. DOCUMENT DATE (YYMMDD)** 27 August 1997

3. DOCUMENT TITLE TEST SET, ELECTRONIC SYSTEMS, TS-3895A/UV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9913 (2) AUTOVON 992-9913
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:03Z
Check the source to verify that this is the current version before use.